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Influence of an Al-Doped ZnO Interface Layer on the Thermochromic Properties of VO_2 Thin Films Deposited by RF Magnetron Sputtering

*Eduard Llorens Balada, Evgeniy Shkondin, Stela Canulescu, and Eugen Stamate**

Vanadium dioxide (VO_2) is intensively studied due to its metallic-to-insulator phase transition exhibiting thermochromic properties. The deposition of thermochromic VO_2 thin films by magnetron sputtering is strongly dependent on process and substrate parameters. Since transparent and conducting electrodes, such as Al-doped ZnO, are widely used for optoelectronic applications, it is highly relevant to investigate the growth of VO_2 thin films on this interface layer with glass as the main substrate. Following this aim, the VO_2 deposition on Al-doped ZnO reveals a faster growth mechanism entailing (020)-oriented films, in detriment of the typical (011)-oriented VO_2 observed when grown directly on glass, with a direct impact on the metal-to-insulator transition. The annealed VO_2 on Al-doped ZnO structure is found to exhibit improved recovery of the thermochromic properties with respect to the VO_2 on glass configuration. Furthermore, Zn diffusion into the VO_2 layer, accompanied by an increased transition temperature and a higher sensitivity to the supplied oxygen flow, is observed. It reveals that the Al-doped ZnO interface layer not only serves as a transparent and conducting electrode but also plays an important role in the functional properties of VO_2 films.

1. Introduction

Vanadium dioxide (VO_2) is attracting significant attention due to its thermoelectric and thermochromic properties. The thermochromic effect is due to the thermally induced and reversible phase change of the crystal structure that occurs as a metal-to-insulator phase transition (MIT) temperature at around 68 °C.[1] Below the MIT temperature, VO_2 presents a monoclinic crystalline structure, while above the MIT temperature, VO_2 exhibits a tetragonal unit cell, also known as the rutile phase.[2] The phase change entails different thermal, optical, and electronic properties.[3]

The possibility of manipulating and switching the electronic, optical, and thermal properties of VO_2 renders this metal oxide as a highly promising candidate to be used in batteries,[4–6] bolometers,[7–15] solar cells or reflectors,[16–20] and smart window applications.[21–28]

The MIT is based on the transition between a rutile (R) phase, whose V−V chains are linear, thus entailing a high conductivity, and the monoclinic (M) phase that possesses zigzag chains because of a distorted and doubled crystalline cell.[29] The metallic behavior exhibited by the rutile VO_2 (R), whose unit cell presents a V atom in orthorhombically distorted octahedral coordination surrounded by six oxygen atoms, is defined by the d−orbital shared electrons placed along the c−axis between the metal atoms through this direction.[30] When the distance between V−V dimers is below a critical value, the 3d orbital electrons are delocalized, resulting in a conductive behavior.[31] Once VO_2 is cooled below its phase transition temperature, the vanadium atoms are displaced in the V−V direction, resulting in an increase in the distance between dimers and a doubling of the number of atoms in the unit cell. The larger distance between V−V pairs and the distorted monoclinic lattice give rise to a 0.7 eV band gap of VO_2 (M), which exhibits an insulator behavior.[32] Although VO_2 has been studied for many years due to its tunable optoelectronic properties, the reason behind its transition temperature remains unknown.[17,30,33–35] Due to its high potential for applications, VO_2 has been studied in recent years employing different deposition techniques. Within the category of chemical deposition techniques, notable success has been achieved using atomic layer deposition,[36–38] chemical vapor deposition,[39,40] and sol–gel processes.[41–43] Various physical deposition methods, including sputtering,[44–46] evaporation,[47–49] molecular beam epitaxy,[50,51] and pulsed laser deposition,[52–55] showed promising outcomes as well. Among these methods, sputtering provides several advantages, such as high process parameter tunability, high compatibility with

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postdeposition processes, and the ability to deposit high-quality thin films over larger areas compared to other methods.[56] Magnetron sputtering of VO_2 can be performed either by means of reactive sputtering using a pure vanadium target[57–61] or using an oxide vanadium compound target, such as V_2O_3 targets,[62,63] VO_2 targets,[64,65] and V_2O_5 targets.[62,66–69] The influence of various sputtering parameters on the microstructure, composition, and resulting VO_2 MIT has been widely studied, and it concluded that the substrate temperature should be between 300 and 600 °C to promote the crystallization of vanadium oxide compounds. Alternatively, annealing needs to be carried out to ensure a crystal quality exhibiting a considerable MIT,[70,71] while the use of O_2 gas flow is important to obtain stoichiometric VO_2.[72–74] Nevertheless, the study of vanadium oxide compounds is challenging due to the presence of several oxidation phases from different vanadium oxide compounds. Therefore, the fine tuning of the oxygen gas flow is regarded as essential.[75–77] Since the MIT of VO_2 is around 68 °C, extensive effort has been devoted to reducing the transition temperature by employing different approaches using, for example, doping with high oxidation state metals, such as W or Ti,[60,78–82] or by epitaxial growth on substrates with lattice parameters close to VO_2.[81,83,84] Moreover, promotion of oxygen vacancies was proven to be an efficient method to decrease the transition temperature in the case of substoichiometric vanadium oxide.[85–87] When used in dedicated devices, VO_2 is only one of the thin films interfacing additional functional layers. Considering the case of smart windows, the thermo- or electrochemically active layer of VO_2 will interface with a transparent conducting electrode (TCE) deposited on a glass substrate. Since vanadium oxides exhibit enhanced intercalation properties and fast solid-state diffusion,[88] it is important to know if the optimized properties of the MIT are preserved or not when depositing the VO_2 thin film directly on the TCE. Aluminum-doped zinc oxide (AZO) is regarded as one of the most promising TCEs due to its very good optoelectronic properties (resistivity below $5 \times 10^{-4}\ \Omega$ cm and transmittance above 80%) and abundance of Zn and Al.[89] Therefore, this study aims to clarify the influence of an AZO interface layer on the properties of radio frequency (RF) magnetron-sputtered VO_2 thin films, in the context of smart windows and other optoelectronic applications.[90]

2. Results and Discussion

VO_2 thin films were deposited on glass wafers varying the oxygen flow from 0 to 10 sccm while keeping the substrate temperature at 400 °C. High-resolution X-ray photoelectron spectroscopy (XPS) was performed to determine the percentage of the oxidation states present in the deposited vanadium oxide films while the carbon and oxide layer formed on top of the sample was firstly removed by an etching step.

As presented in **Figure 1** ((a) for 0.5 sccm and (b) for 1 sccm oxygen flow, respectively), oxide states were assigned to each deconvoluted peaks, according to previous reported studies: V_2O_5 corresponding the largest binding energy range from 516.8 to 518 eV,[91] VO_2 from 515.8 to 516.7 eV,[76] V_2O_3 from 515.7 to 515.10 eV,[76] and VO from 513.9 to 512.70 eV.[92] The area of each given peak was then compared to the O_{1s} peak area to determine the measured composition of vanadium oxide and the one corresponding to each oxidation state, that is, the composition of each vanadium oxide compound.[76]

The influence of the oxygen flow on the VO_2 thin film properties is presented in **Figure 2** as following, (a) the composition of vanadium oxide compounds by X-ray diffraction (XRD), (b) Raman spectra, (c) XRD spectra, and (d) the temperature-dependent transmittance. Regardless of the oxygen flow, VO_2 was found in all films, being the most predominant vanadium oxide for oxygen flows of 1 sccm or higher. Flows lower than 1 sccm promoted the formation of a mixture of vanadium oxide compounds, with the V_2O_3 phase as predominant. Independently of the oxygen flow rate, a mixture of vanadium oxides with different oxidation states was found in each film. The characteristic Raman shift found at around 141 cm^{-1} (see Figure 2b), which can be attributed to V^{5+}—O stretching mode of terminal oxygen ions or to the M1 phase of VO_2,[75] is highlighted in the vanadium oxide films deposited between 0.5 and 2 sccm of oxygen flow and decrease in intensity for the films deposited outside of this range.[75]

These results indicate that a higher crystallinity is promoted for films deposited at low oxygen flows. When no oxygen or higher oxygen flow rates than 2 sccm were used, the intensity of Raman spectra peaks decreased, implying lower crystal quality. Raman bands found at 282 cm^{-1}, and 405 cm^{-1} are related to

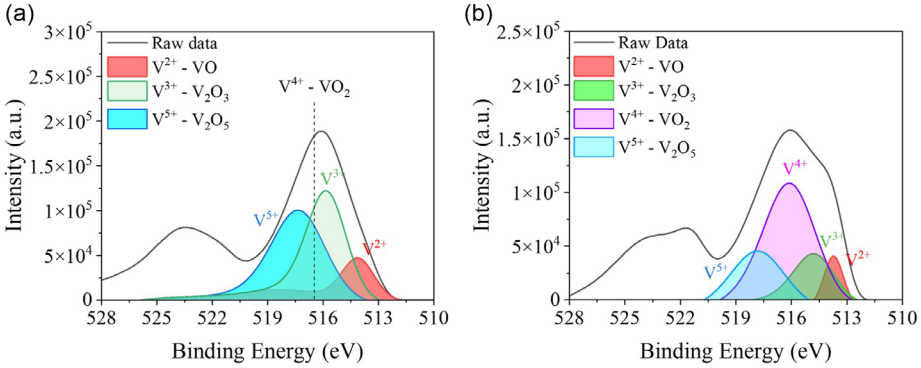


Figure 1. V_{2p} high-resolution XPS pattern deconvoluted to study the oxidation states present in the vanadium oxide thin film deposited using an oxygen flow rate of a) 0.5 sccm and b) 1 sccm.

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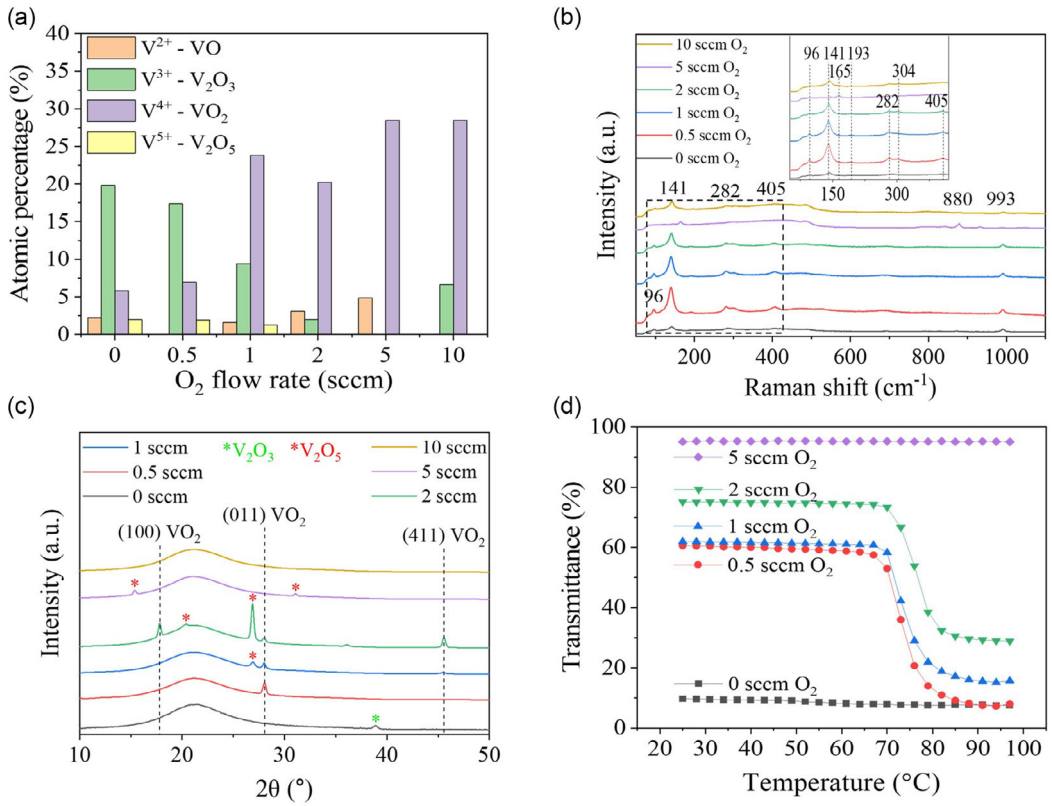


Figure 2. Effect of the oxygen flow rate on the vanadium oxide deposited thin films using a glass substrate on the a) oxidation states present in vanadium oxide films characterized by XPS, b) Raman spectra of the corresponding vanadium oxide thin films, inlet shown from 50 to 410 cm^{-1} of the Raman shift, c) XRD patterns of vanadium oxide thin films, and d) temperature-dependent transmittance obtained at 2000 nm wavelength.

the formation of V_3O_7 phases.[75] The Raman band observed at 880 cm^{-1} is formed due to VO_2 stretching modes.[93] Noticeably, none of the characteristic Raman peaks of the M1 phase of VO_2 expected at 224, 393, and 612 cm^{-1} have been detected[75] while the peak at 192 cm^{-1} was very weak. However, this finding is consistent with other reported results on VO_2 deposition on ZnO by magnetron sputtering[94] and on Al_2O_3 by electron beam[95] where Raman spectra exhibiting very few peaks have been reported in certain ranges of substrate temperatures or substrate bias. In this regard, XPS and Raman are useful characterization techniques but cannot determine whether the vanadium oxide films are able to show thermochromic properties.

For a better understanding of the crystal quality, XRD was employed, and the patterns of the films are presented in Figure 2c. Films deposited with oxygen flow rates between 0.5 and 2 sccm exhibited the characteristic (011) VO_2 peak at the 2θ angle of around 28°, indicating the formation of the thermochromic VO_2 phase. These results are consistent with the Raman spectra of Figure 2b. In contrast, the film deposited without oxygen exhibited a (006)-oriented V_2O_3, suggesting a metallic behavior, as the V_2O_3 transition temperature is around 160 K.[96] This agrees with the dominant V^{3+} valence state derived from the XPS data (see Figure 2a). A higher oxygen flow led to the formation of V_2O_5 or amorphous films, suggesting poor thermochromic properties.

The transmittance of these films was measured at different temperatures. The temperature-dependent transmittance values at the wavelength of 2000 nm are shown in Figure 2d as a function of the oxygen flow during the deposition process. Films deposited with an O_2 flow between 0.5 and 2 sccm revealed the desired MIT, with the films deposited at the lowest oxygen flow presenting an optical transmittance difference exceeding 50%. These results are consistent with the XRD (011) VO_2 peak and Raman shift found at 141 cm^{-1}. When no O_2 flow is used, the metallic V_2O_3 phase becomes predominant, while a high oxygen flow promotes a dielectric phase that does not show an MIT, thus highlighting the importance of finely tuning the O_2 flow during the deposition process. The films that exhibited V_2O_5 predominant phases or amorphous-like structures, deposited at a high oxygen flow rate, showed similar temperature-dependent transmittance patterns, revealing an insulator behavior through all the temperature ranges.

2.1. VO_2 Deposition on AZO Interface Layer

To produce smart windows, one must use a TCE, which in this study was an interface layer of AZO. The XRD patterns of the stack of VO_2/glass and VO_2/AZO/glass films (0.5 sccm O_2 flow rate) and the temperature-dependent transmittance

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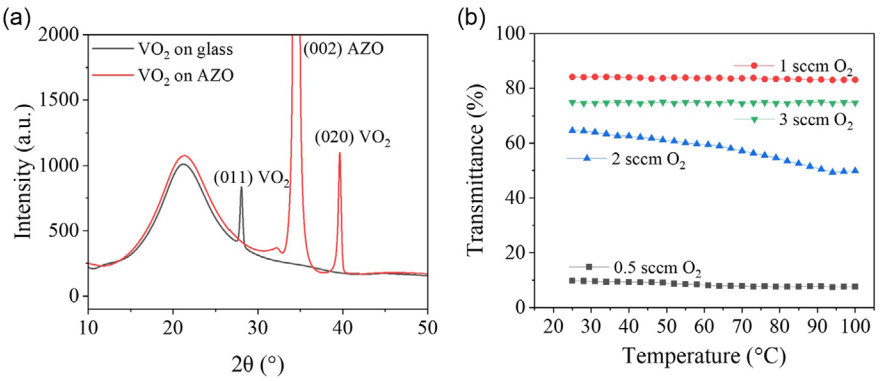


Figure 3. a) XRD patterns of VO_2 deposited on glass and AZO. b) Temperature-dependent transmittance measured at 2000 nm wavelength of VO_2/AZO/glass stack.

for the VO_2/AZO/glass for different O_2 flows are shown in **Figure 3**a,b, respectively.

When deposited on AZO/glass, the characteristic (011) VO_2 peak, typically observed at 2θ of around 28°, shifted to (020) orientation (centered at 2θ around 39.5°). This change has been reported before, but most of those studies involved sapphire wafers or a post-annealing treatment.[97,98] Although the (020) intensity peak was higher than the one observed for the (011) peak, implying highly oriented VO_2 films on AZO, the MIT almost vanishes, resulting in a flattened transition range. In contrast, a bare glass substrate led to a transition change of around 50% of transmittance (see Figure 2d). The occurrence of either a flat transition or MIT could be explained by a different growing mechanism of the (020) orientation that might exhibit an increased deposition rate that leads to higher stress of the VO_2 films.[99,100] Moreover, the MIT transition is also dependent on the grain size, grain boundaries, crystal imperfections, and voids.[101]

2.2. High-Temperature Growth of VO_2

To further optimize the sputtering parameters seeking the formation of stoichiometric VO_2 films, the following depositions were carried out at 450 °C to enhance the crystallinity. The O_2 flow was increased in small steps to better capture the compositional variations. The thermochromic properties are presented in **Figure 4**a as deposited and (b) after annealing in N_2 atmosphere at 500 °C, respectively.

The deposition time was reduced to 2000 s to decrease the thickness and the possible stress while varying again the oxygen flow. Figure 4a reveals the importance of fine tuning of the O_2 flow rate to achieve stoichiometric VO_2, which in this case was found to be 0.4 sccm. Furthermore, by annealing the samples under the N_2 atmosphere for 1 h, the vanadium oxide deposited at low oxygen flow could recover the MIT, as shown in Figure 4b. The samples deposited at oxygen flows expected to provide stoichiometric VO_2 could not recover their MIT and, even more, revealed an increase in their insulator behavior over the entire range of temperature, with a higher transmittance than before the annealing (see Figure 4b).

To compare the substrate influence on the thermochromic properties of VO_2, an AZO/glass of $1.5 \times 1.5\ cm^2$ and a bare glass (without AZO) of $1.5 \times 1.5\ cm^2$ substrates were placed on the substrate holder as to provide the very same sputtering conditions.

Figure 5 reveals the comparison of using both substrates under the same deposition conditions (a) as deposited and (b) after annealing, respectively.

Figure 5a shows the importance of substrate choice for VO_2 thermochromic applications. While the VO_2 film deposited on

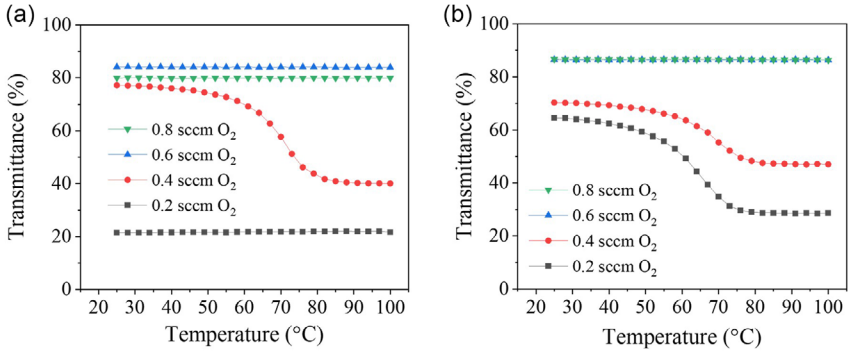


Figure 4. a) Temperature-dependent transmittance measured at 2000 nm wavelength of VO_2/AZO/glass stack, deposited at 450 °C during 2000 s. b) Temperature-dependent transmittance measured at 2000 nm wavelength of annealed VO_2/AZO/glass stack under N_2 atmosphere for 1 h at 500 °C.

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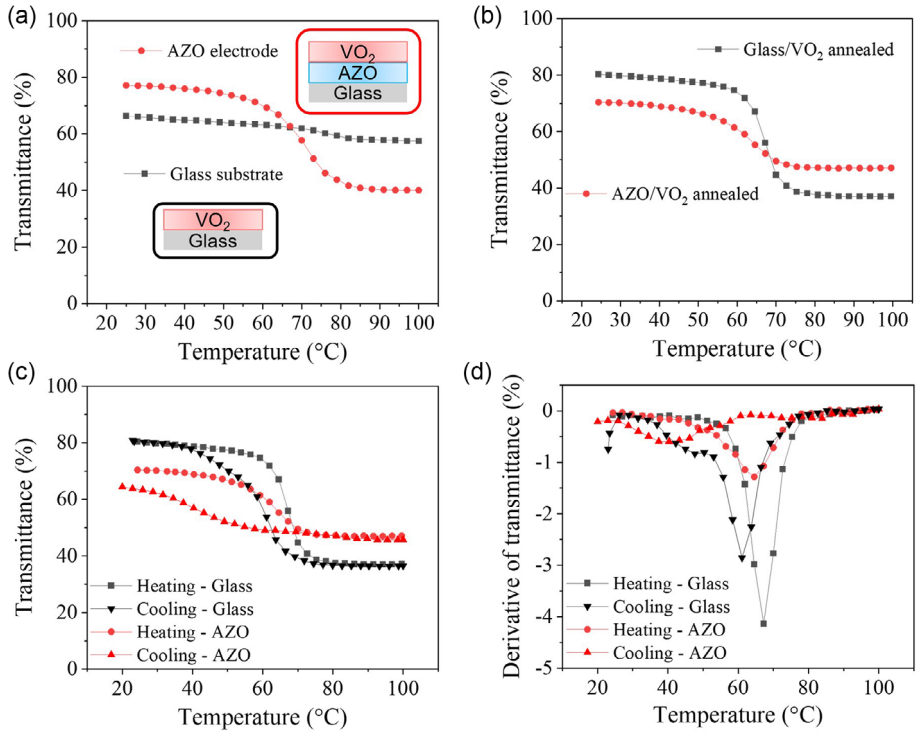


Figure 5. a) Comparison between the thermochromism of VO_2 deposited on a glass versus AZO/glass stack. b) Temperature-dependent transmittance at 2000 nm wavelength of annealed VO_2 deposited on glass and AZO/glass stack. c) Hysteresis loop of temperature-dependent transmittance of both annealed configurations. d) The first derivative of heating and cooling for each stack of layers.

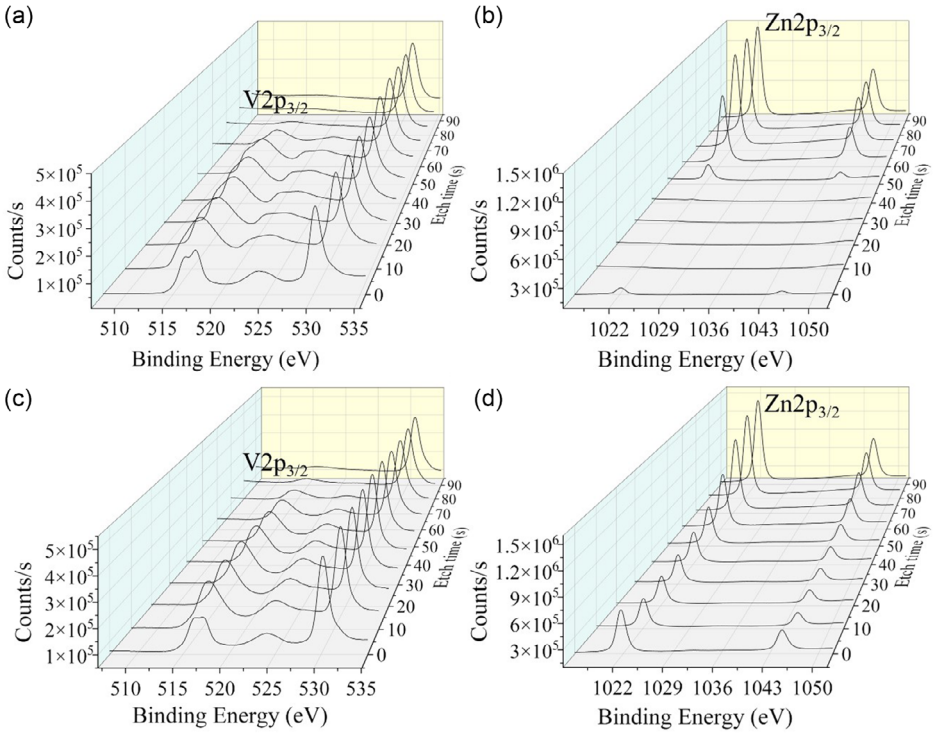


Figure 6. XPS depth profile spectra of as-deposited VO_2/AZO/glass stack a) V $2p_{3/2}$ peak, b) Zn $2p_{3/2}$ peak, and annealed VO_2/AZO/glass stack c) V $2p_{3/2}$ peak, and d) Zn $2p_{3/2}$ peak.

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AZO/glass shows an optical difference between the insulator and metallic state, the VO_2 film deposited on glass shows a small decrease over the entire range of temperatures. However, once annealed, as shown in Figure 5b, the VO_2 deposited on glass recovered the MIT and exhibited a higher optical difference with respect to the VO_2/AZO/glass stack. The hysteresis loops depicted in Figure 5c reveal an irreversible hysteresis of the VO_2/AZO/glass stack, while for the VO_2/glass stack, the process is reversible with a small gap between curves. Figure 5d shows at which temperature the MIT takes place by measuring the derivative of the transmittance. Upon heating, the MIT of the VO_2/glass stack is found around 68 °C and, upon cooling, at 61 °C, indicating a small unwanted hysteresis width. The VO_2/AZO/glass stack revealed a minimum of the first derivative, hence MIT temperatures of 64 and 40 °C, upon heating and cooling, respectively. Therefore, the annealed VO_2/AZO/glass stack produces a thermochromic effect of less quality. To understand the reason for this degradation, XPS depth profiles of the V $2p_{3/2}$ and Zn $2p_{3/2}$ were measured for (a) and (b) as-deposited films and (c) and (d) after annealing, respectively.

Figure 6a,b reveal a considerable intermixing layer where Zn and V coexisted at 50, 60, and 70 s of etching time. Therefore, these results indicate that Zn can diffuse for a certain depth into VO_2, compromising the thermochromic properties, as shown in Figure 5c. After annealing, the XPS depth profile revealed a measurable level of Zn through entire the VO_2 layer as presented in Figure 6d. The increased Zn detected at the surface of each sample can be attributed to a Zn cross contamination process.[102]

Consequently, the reason for the non-reversibility of the hysteresis loop of the VO_2/AZO/glass stack can be correlated with the diffusion of Zn originating from the AZO interface layer within the VO_2 layer. A possible solution would be to increase the thickness of the VO_2 layer further, so that the influence of the interface layer gets reduced, or to use a capping layer to prevent Zn diffusion into the thermochromic layer.

3. Conclusion

RF sputtering has been employed to deposit VO_2 from a VO_2 target on glass and on AZO precoated glass substrates. The results show the high sensitivity of vanadium oxide to be grown with different oxidation states, leading to the formation of different vanadium oxides. This understanding was used to produce a VO_2/AZO/glass stack of layers for smart window applications. As a result, the growth of VO_2 on AZO/glass substrates exhibited a different phase orientation, hence a different growing mechanism, and a different deposition rate. Therefore, this indicates that one must tune separately the VO_2/AZO/glass stack to obtain adequate thermochromic properties. A postdeposition annealing process under N_2 flow improved the crystal quality and the VO_2 thermochromic properties either deposited on glass or AZO/glass stack. Despite this improvement, the VO_2/AZO/glass stack revealed an increase in the MIT temperature and a large width irreversible hysteresis loop due to Zn diffusion towards the VO_2 layer. Further research should consider the need to use a capping layer between the VO_2 layer and the AZO layer to avoid Zn contamination.

4. Experimental Section

VO_2 was deposited on fused silica glass wafers by RF magnetron sputtering from a 3-inch VO_2 target using a Kurt J. Lesker 2017 PRO Line PVD75 thin film deposition cluster system. The AZO thin films were deposited from a target of 2% Al_2O_3 to 98% ZnO by weight, using the same setup. Both targets (VO_2 and AZO) were supplied by Kurt Lesker. The pressure and power were optimized to 3 mTorr and 100 W, respectively. The distance from the target to the substrate was 15 cm. Argon flow rate was set to 50 sccm while the oxygen flow rate was varied between 0 and 10 sccm. The substrate temperature varied between 400 and 450 °C and the deposition time between 2000 and 6000 s.

AZO thin films were RF sputtered using the same deposition setup with the following optimized conditions: 200 °C substrate temperature, 2 mTorr discharge pressure, 140 W RF power, and 2200 s deposition time, which resulted in a 110 nm thick AZO layer on top of 4-inch glass wafers that were cut after deposition into 1.5×1.5 cm^2 pieces.[103–105]

Annealing was performed in a quartz tube under N_2 atmosphere for 1 h at 500 °C, with a heating and cooling rate of 3 °C s^{-1}.

XRD spectra were obtained using a Rigaku Smartlab XRD system equipped with an automated moving stage. Raman spectra were taken using an inVia Raman microscope from Renishaw with a monochromatic green source laser using 0.05 W of power at a wavelength of 532 nm.

Thickness was measured using an Ellipsometer VASE from J.A. Woollam. Optical transmission was obtained using a UV-Vis-NIR CRAIC 20/30 PV microspectrophotometer. A Linkam PE120 heating stage was used to record the transmittance as a function of temperature.

K-alpha X-ray photoelectron spectrometer (Thermo Fisher Scientific) was used to study the elemental composition of the films. A monochromatic Al-Kα X-ray source with a spot size of 400 μm was used. A survey spectrum was obtained from 0 to 1350 eV, using a pass energy of 200 eV, 2 scans, and a dwell time of 50 ms. VO_{2p}, Zn_{2p}, O_{1s}, Al_{2p}, and C_{1s} peaks were obtained using a pass energy of 50 eV, 5 scans, and a dwell time of 50 ms. The C_{1s} peak was used as a reference peak at 284.8 eV. The Thermo Fischer Avantage Software was employed to determine the composition and peak fitting using 100 maximum iterations of a product of a Gauss–Lorentz mix to fit all the levels. An Ar ion gun was used to etch 10 layers using an ion energy of 1000 eV for 10 s to characterize the composition of the VO_2 thin films through depth profiling. The depth profiles were obtained using an Ar ion gun at 1000 eV ion energy for 10 s between the 10 etched levels.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Eduard Llorens Balada: conceptualization (lead); data curation (lead); formal analysis (lead); investigation (lead); methodology (lead); validation (equal); visualization (lead); writing—original draft (lead). **Evgeniy Shkondin**: conceptualization (supporting); methodology (supporting); validation (supporting). **Stela Canulescu**: methodology (supporting); validation (supporting); writing—review and editing (supporting). **Eugen Stamate**: conceptualization (supporting); formal analysis (supporting); methodology (supporting); supervision (lead); validation (supporting); writing—review and editing (equal).

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

aluminum-doped zinc oxides, interface layers, magnetron sputtering, thermochromic thin films, vanadium dioxides

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